

188.429 Business Intelligence (VU 4.0) – 2025W

Assignment 1 Report

Group 034

Student A: Kadic Amina

Student B: Kasapovic Meliha

Title: Dimensional Modelling and ETL implementation

1. Synthetic tables (Table_X and Table_Y)

To extend OLTP snapshot, two synthetic tables were created: table `tb_project` (as Table_X) and `tb_projectemployee` (as Table_Y).

Table_X - `tb_project`

This entity table contains information about projects. It contains 6 rows and the following columns:

- `id`: INT NOT NULL PRIMARY KEY - contains unique project id
- `project_name`: VARCHAR(100) NOT NULL - contains name of the project
- `budget`: NUMERIC(12,2) - contains budget provided for the project
- `goal`: TEXT - contains goal of the project
- `sponsor`: TEXT - contains if the project is funded by government or non-government organization or both
- `start_date`: DATE - contains start date of the project
- `end_date`: DATE - contains end date of the project

This table provides information about project that could help with future analysis of resources and project outcomes.

Table_Y - `tb_projectemployee`

This is a bridge table that creates many-to-many relationship between projects and employees. It contains 12 rows and the following columns:

- `id`: INT NOT NULL PRIMARY KEY - primary key of this table
- `project_id`: INT REFERENCES `tb_project(id)` - contains `project_id` on which employee works
- `employee_id`: INT REFERENCES `tb_employee(id)` - contains id of an employee that works on the project
- `hoursworked`: NUMERIC - contains information on number of hours that the employee spent working on this project

This bridge table connects entity table `tb_project` with the existing OLTP table `tb_employee`. It contains information that can improve analysis of staff allocations and workload distributions.

2. Business and Analytic Questions

Student A

- Q1:** Which sensors and location types (urban, industrial, suburban) achieve the lowest average `data_quality_score` and require additional calibration?
- Q2:** How does the average concentration of a certain parameter change over time within individual projects (`dim_project`)?
- Q3:** Are there cities or projects that have recorded high levels of pollution while simultaneously having a low data quality score?
- Q4:** Which projects record the highest number of days with an average exceedance of a certain parameter above a defined threshold?

Student B

- Q1:** Which service types (`dim_servicetype`) generate the highest total costs (`servicecost`) per project?
- Q2:** Is there a difference in the average service duration (`durationminutes`) between technicians of different qualification levels?
- Q3:** In which cities are services most often performed and does the frequency of services differ by sensor type?
- Q4:** How does the quality of performed services (`servicequality`) vary between different technicians and projects?

3. Star Schema Diagram

Exported ER diagram of our star schema from DBeaver.

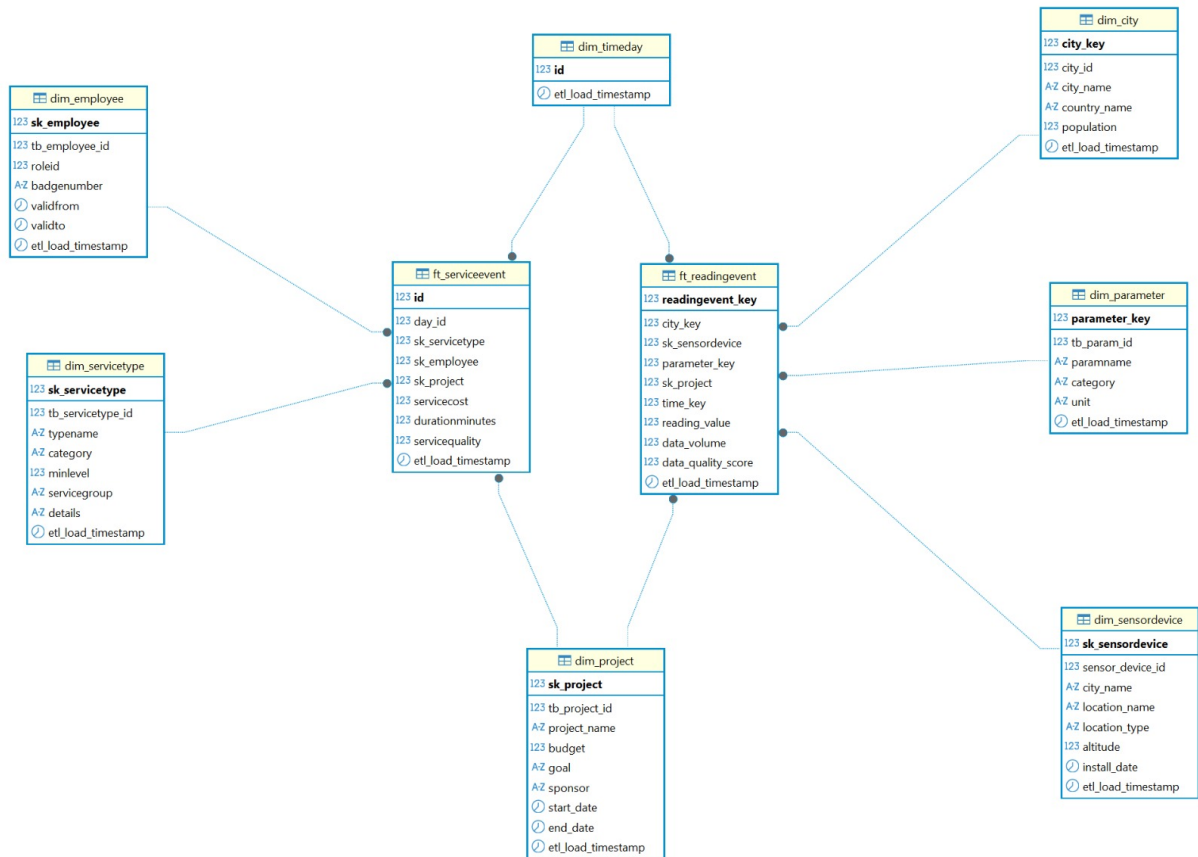


Figure 1: Star Schema ER Diagram (dwh.034)

4. Fact Tables

Fact 1 (ft_readingevent) – Student A

Responsible student: Kadic Amina

Business motivation: This fact table enables the analysis of daily air quality measurements and operational indicators of data collection. It is possible to assess pollution patterns, quality of collected data, and quantity of telemetry data by device, city, and date.

Definition of grain: Each row represents a single reading of a particular environmental parameter by a particular device on a particular day.

Measures and aggregation rules:

- **reading_value** – measured value of the environmental parameter (AVG)
- **data_volume** – amount of transferred data (SUM, fully additive)
- **data_quality_score** – quality score of collected data (AVG)

Related dimensions:

- **dim_timeday** – measurement date
- **dim_sensordevice** – the device that performed the measurement
- **dim_parameter** – environmental parameter to be measured
- **dim_city** – geographic location (conformal dimension divided by ft_serviceevent)
- **dim_project** – the project within which the reading was recorded

Fact 2 (ft_serviceevent) – Student B

Responsible student: Kasapovic Meliha

Business motivation: This fact table records all service events related to sensor maintenance and diagnostics. It allows for analysis of costs, duration and quality of service interventions and monitoring technician performance and maintenance frequency by type of service or device.

Definition of grain: One row represents one service event that a technician performed on a specific device on a specific date.

Measures and aggregation rules:

- **servicecost** – total service cost (SUM, fully additive)
- **durationminutes** – service duration in minutes (AVG)
- **servicequality** – evaluation of the quality of the performed service (AVG)

Related dimensions:

- **dim_timeday** – service execution date
- **dim_servicetype** – type of service activity (e.g. calibration, diagnostics)
- **dim_employee** – the technician who performed the service
- **dim_project** – the project within which the service was performed

Common dimensions: **dim_timeday** and **dim_project** are conformal (shared) dimensions between fact tables **ft_serviceevent** and **ft_readingevent**.

5. Dimension Tables

Seven dimension tables were implemented to create star schema. Each dimension table was implemented to help with answering on business and analytical questions.

dim_timeday

This dimension table provides time information for all readings and service activities. It stores only id, which contains the distinct calendar days observed in the dataset. It is shared dimension used by both fact tables to provide time aggregation and comparison. **Hierarchy:** Single level (day of the year) **Columns:** id (PK), etl_load_timestamp. **Granularity choice:** Only one time dimension.

dim_city

This dimension stores geographic context for monitoring stations and devices. **Hierarchy:** Country → City. **Columns:** city_key (PK), city_id, city_name, country_name, population, etl_load_timestamp **Notes:** Shared by ft_readingevent to support geographical aggregations and comparisons

dim_sensordevice

This dimension describes each installed device and its geographical characteristics. **Hierarchy:** City → Location → Device. **Columns:** sk_sensordevice (PK), sensor_device_id, city_name, location_name, location_type, altitude, install_date, etl_load_timestamp **Usage:** Linked to ft_readingevent to analyze sensor performance

dim_parameter

This dimension defines parameters measured by devices and their units. **Hierarchy:** Category → Parameter. **Columns:** parameter_key (PK), tb_param_id, paramname, category, unit, etl_load_timestamp. **Usage:** Linked to ft_readingevent for breakdowns by pollutant type.

dim_project

Created from the entity Table_X (tb_project). This table adds managerial and financial context for projects. **Hierarchy:** Project Sponsor → Project **Columns:** sk_project (PK), tb_project_id,

project_name, budget, goal, sponsor, start_date, end_date, etl_load_timestamp. **Usage:** Linked to both fact tables to analyze service costs and data quality for projects.

dim_employee

Represents employees performing service events. Since employees can change roles, this table supports historical tracking if extended with SCD Type 2. **Hierarchy:** Role → Employee. **Columns:** sk_employee (PK), tb_employee_id, roleid, badgenumber, validfrom, validto, etl_load_timestamp. **Usage:** Linked to ft_serviceevent. SCD Type 2 is not fully implemented, but history can be tracked using columns validfrom and validto.

dim_servicetype

Contains metadata describing maintenance. **Hierarchy:** Service Group → Category → Service Type **Columns:** sk_servicetype (PK), tb_servicetype_id, typename, category, minlevel, servicegroup, details, etl_load_timestamp. **Usage:** Linked to ft_serviceevent and used for analyzing service quality and cost by service category and qualification level.

Special Dimensions and Design Choices

Only one time dimension was implemented since both fact tables operate on daily granularity. No explicit degenerate dimension was used and all identifiers were handled with surrogate keys in fact tables. No junk dimension was needed because all categorical attributes were normalized into proper dimensions.

6. Snowflake vs. Star

In this project, as per the instructions a star schema was applied and the dimension tables are directly connected to the central fact tables. As opposed to that, a snowflake schema would allow a higher degree of data normalization and reduce redundancy since the dimensions themselves would be interconnected, and would establish the hierarchy easier- e.g city and country. However, this schema would increase the complexity of the database structure and would require more joins which could negatively impact the performance of queries, in this case the snowflake schema would perhaps bring more complexity than advantages.

7. ETL and Validation Summary

During the Extract–Transform–Load process, checks were performed to verify the accuracy and consistency of the data in the star schema `dwh_034`.

- **Row count check:** The row count in staging tables was compared to dimension and fact tables in data warehouse. The row counts match, confirming that the data transfer was complete.
- **Null value check:** The presence of NULL values in the key columns of the fact tables `ft_readingevent` and `ft_serviceevent` was checked. The result shows that no foreign key is NULL.
- **Data quality check :** The minimum and maximum ranges of measurement values and operational indicators were analyzed. All values are within logical and expected limits (no negative values or scores outside the 1–5 range).
- **Referential integrity check :** It was checked that all references between fact tables and their dimensions exist. Result: 0 missing keys - all relationships are correct.
- **Consistency check :** The average values of the main measures were calculated to check the numerical consistency. The average values obtained are realistic and confirm no extreme deviations.
- **Time coverage check :** The range and total number of days in the time dimension were checked. The `dim_timeday` dimension covers all dates present in the fact tables, without gaps in the time series.

- **City distribution check :** The number of readings per city was analyzed to check the correct relationship between `ft_readingevent` and `dim_city`. The distribution of readings across cities is consistent.

8. Reflection and Lessons Learned

Student A – Kadic Amina During this assignment, I learned how to transform OLTP transactional data into analytically suitable structures that allow decision making. I learned how to design a star schema and how to properly define grain in fact tables, and the importance of conforming dimensions of fact tables in order to connect different facts. This process taught me how data cleansing, validation, and referential integrity are important for dimensional modeling and how they are applied in a real business intelligence context.

Student B – Kasapovic Meliha This project provided me with deeper understanding of principles of data warehousing and the ETL process, from extraction from the staging schema to creating a dimensional model ready for analytics. I learned the most about aggregation logic, connecting fact tables with common dimensions, and practical data quality checks. Validation through SQL checks showed me how important it is to systematically test each phase to avoid errors in future analyses. With this practical work, I saw how much automation and clearly structured ETL steps help with repeatability and reliability of the entire system.